

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
81 Higuera Street, Suite 200
San Luis Obispo, California 93401**

**WATER RECYCLING REQUIREMENTS ORDER NO. 01-083
Waste Discharger Identification No. 3 271018001**

For

**MONTEREY COUNTY SANITATION DISTRICT NO. 388,
CARMEL VALLEY RANCH ZONE NO. 1
PRODUCER OF RECYCLED WATER
FOR
CARMEL VALLEY RANCH GOLF COURSE
Monterey County**

The California Regional Water Quality Control Board, Central Coast Region (hereafter Regional Board), finds:

FACILITY OWNER AND LOCATION

1. Monterey County Sanitation District No. 388 (hereafter Producer) owns a domestic wastewater treatment plant (hereafter Facility) located on the Carmel Valley Ranch's restricted access golf course, 3/4 mile east of Robinson Canyon Road, in central Carmel Valley. The facilities are located six (6) miles east of Carmel, as shown on Attachment "A" of this order. Carmel Valley Ranch operates the treatment plant under contract with the Discharger.

PURPOSE OF ORDER

2. On September 21, 1999 the Discharger submitted a Report of Waste Discharge for authorization to continue producing recycled water for the irrigation of Carmel Valley Ranch Golf Course. This Order continues and updates water recycling requirements that are intended to allow the discharge described in the Discharger's Report of Waste Discharge, and uphold State water quality standards.

SITE/FACILITY DESCRIPTION

3. **Collection System** - The Facility has approximately five (5) miles of sanitary sewer lines and two (2) lift stations. Each lift station is equipped with backup pumps, standby power, and high water alarms.
4. **Treatment** - Up to 100,000 gallons-per-day (gpd) of domestic wastewater are treated at the Facility. The Facility's tertiary treatment process consists of an equalization flow basin with aeration systems, a small rotary screen for removal of influent debris, rotating biological contactors, secondary clarifiers, gravity sand filters, a chlorine contact tank, flow monitoring, and chemical feed systems. Auxiliary systems include an aerobic sludge digester, sludge dewatering system, and backup power generation.
5. **Design and Current Capacity** - The treatment plant design criteria and current flows, in gallons-per-day (gpd), are as follows:

	Flow (gpd)	
	Design	Current
30-day average dry weather flow	100,000	75,000
Maximum daily flow	150,000	100,000

6. **Solids Disposal** - Solid wastes generated by the Facility (primarily digested and dewatered sludge) are disposed by truck at the Monterey Regional Waste Management District disposal site at Marina.
7. **Recycled Water Storage and Use** - Recycled water is discharged to two (2) ponds adjacent to the Facility (as shown on Attachment "A"). Recycled water is pumped from the ponds, blended with well water in two (2) golf course lakes located on the golf course, and spray-irrigated throughout the golf course. All storage facilities are lined to prevent seepage to ground water. For most of each year, recycled water comprises approximately ten (10) percent of the overall golf course water use. In winter months, recycled water may comprise 100 percent of overall golf course water use. During extended rainy periods, when there is no demand for recycled water, the excess recycled water flows may be discharged to a storage basin that is also used as a golf driving range.
8. **Geology** - The golf course is located on gently rolling topography consisting of sandy loam soils to a depth of 80 feet.
9. **Surface Water** - The Carmel River is located adjacent to the golf course recycled water application area and approximately 600 feet northeast of the treatment plant (as shown on Attachment "A"). Carmel River flows west to the Pacific Ocean.
10. **Ground Water** - Depth to ground water in seven (7) monitoring wells ranged from 19 to 35 feet in October 2000. A comparison of 1980 (prior to discharge) and 2000 mineral quality of ground water in monitoring wells No. 3 and 4, downgradient of the recycled water discharge area, is as follows:

Constituent	1980 (prior to discharge)	2000
Total Dissolved Solids	184-336 mg/L	320-350 mg/L
Chloride	24-43 mg/L	32-40 mg/L
Nitrate (as N)	<1.2 mg/L	0.3-0.7 mg/L

11. **Storm Water** - The Facility is not required by U.S. Environmental Protection Agency regulations to obtain a storm water permit and the Board does not regulate the Facility's storm water runoff.

Beneficial Uses

12. Present and anticipated beneficial uses of groundwater in the vicinity of the recycled water area include:
 - a. Domestic and Municipal Supply;
 - b. Agricultural Supply; and,
 - c. Industrial Supply.
13. Present and anticipated beneficial uses of the Carmel River in the vicinity of the recycled water area include:
 - a. Municipal and Domestic supply;
 - b. Agricultural supply;
 - c. Industrial service supply;
 - d. Ground water recharge;
 - e. Water contact recreation;
 - f. Non-contact water recreation;
 - g. Wildlife habitat;
 - h. Cold freshwater habitat;
 - i. Fish Migration (including steelhead trout); and,
 - j. Fish Spawning.

MONITORING PROGRAM

14. Monitoring and Reporting Program (MRP) 01-083 is a part of the proposed Order. The MRP requires routine influent and effluent monitoring to verify compliance with this order and protection of water quality.

BASIN PLAN

15. The Water Quality Control Plan, Central Coast Basin (Basin Plan) was adopted by the Regional Board on November 19, 1989, and approved by the State Board on August 16, 1990. The Regional Board approved

amendments to the Basin Plan on February 11, 1994, and September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters.

CEQA

16. These Water Recycling Requirements are for an existing facility and are exempt from the provisions of the California Environmental Quality Act (Public Resources Code, Section 21000, et. seq.) in accordance with Section 15301, Article 19, Chapter 3, Division 6, Title 14 of the California Code of Regulations.

EXISTING ORDERS AND GENERAL FINDINGS

17. The discharge has been regulated by Wastewater Reclamation Requirement Order No. 89-03, adopted by the Board on March 10, 1989. The Regional Board has regulated this discharge since 1980.
18. Carmel Valley Ranch uses all recycled water supplied by the Producer. This primary user of recycled water is regulated by separate water recycling requirements Order No. 01-084.
19. State Department of Health Services' criteria for use of recycled water is in Title 22, Division 4, Chapter 3, of the California Code of Regulations. The Board has consulted with the State Department of Health Services regarding the regulation of this discharge.
20. Discharge of waste is a privilege, not a right, and authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should assure this and mitigate any potential adverse changes in water quality due to the discharge.
21. On June 22, 2001, the Board notified the Discharger and interested agencies and persons of its intent to update water recycling requirements for the discharge and has provided them with a copy of the proposed

order and an opportunity to submit written views and comments.

22. After considering all comments pertaining to this discharge during a public hearing on September 14, 2001, this Order was found consistent with the above findings.

IT IS HEREBY ORDERED, pursuant to authority in Section 13263 of the California Water Code, Monterey County Sanitation District No. 388, Carmel Valley Ranch Zone No. 1, its agents, successors, and assigns, may discharge waste at Carmel Valley Ranch Golf Course providing compliance is maintained with the following:

All technical and monitoring reports submitted pursuant to this Order are required pursuant to Section 13267 of the California Water Code. Failure to submit reports in accordance with schedules established by this Order, attachments to this Order, or failure to submit a report of sufficient technical quality to be acceptable to the Executive Officer, may subject the discharger to enforcement action pursuant to Section 13268 of the California Water Code. The Regional Board will base all enforcement actions on the date of Order adoption.

Notes:

Other prohibitions and conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January 1984. Applicable paragraphs are referenced in paragraph D.3. of this Order.

Throughout these requirements footnotes are listed to indicate the source of requirements specified. Requirement footnotes are as follows:

BP	Basin Plan
Design	Design of Facility
CWC	California Water Code
T22	California Code of Regulations, Title 22, Recycled Water Criteria

Requirements not referenced are based on staff's best professional judgement.

A. PROHIBITIONS

1. Discharge to other than the golf course recycled water areas shown in Attachment "A" is prohibited.^{T22}
2. Discharge of wastes, including overflow, bypass, and overspray from treatment, storage, transport, or recycled water areas to adjacent drainageways or adjacent properties is prohibited.^{T22}
3. Bypass of the treatment facility and discharge of untreated or partially treated wastes directly to the lakes or recycled water areas is prohibited.^{T22}

B. RECYCLED WATER PRODUCER REQUIREMENTS

1. All recycled water produced shall be in conformance with the recycled water criteria established in Title 22, Division 4, Chapter 3, of the California Code of Regulations.
2. Daily flow averaged over each month shall not exceed 100,000 gallons.^{Design}
3. Effluent discharged shall not exceed the following limitations:

Constituent	Units	Monthly 30-day Average	Maximum
Total Dissolved Solids	mg/L	-	1,000
Chloride	mg/L	-	250
BOD ₅	mg/L	10	25
Total Suspended Solids	mg/L	10	25

4. Turbidity of effluent shall not exceed an average of 2 NTU within a 24-hour period; 5 NTU more than 5 percent of the time within a 24-hour period; and 10 NTU at any time.^{T22}
5. The median concentration of total coliform bacteria measured in the disinfected effluent

shall not exceed an MPN of 2.2 per 100 milliliters utilizing the bacteriological results of the last seven days for which analyses have been completed and the number of total coliform bacteria shall not exceed an MPN of 23 per 100 milliliters in more than one sample in any 30 day period. No sample shall exceed an MPN of 240 total coliform bacteria per 100 milliliters.^{T22}

6. Effluent shall be subject to a chlorine disinfection process that provides a CT (chlorine concentration times modal contact time) value of not less than 450 milligram-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow.^{T22}
7. Effluent shall not have a pH less than 6.5 or greater than 8.4.^{BP}
8. Free chlorine residual in effluent shall equal or exceed 1 mg/l, as measured within the chlorine contact zone.
9. A minimum freeboard of 1 foot shall be maintained in the golf course lakes containing recycled water.
10. Delivery of recycled water to a specific user shall cease during any period the user cannot comply with its water recycling requirements. Use of recycled water may resume when the user can once again comply with its water recycling requirements.
11. Water recycling facilities shall be designed and operated to retain all wastewater and recycled water onsite during a 100-year rainfall season.
12. All commercial grease traps at the Facility shall be inspected on a monthly basis and cleaned when appropriate. A log of all cleanings shall be kept and made available upon Executive Officer request.
13. The drainage valve of the golf course driving range storage basin shall be closed and inspected prior to recycled water discharge to the basin. Only stormwater may be released from this basin via the open drainage valve. Stormwater shall be free of any recycled

water so that recycled water is never discharged to the Carmel River. If recycled water must spill into the basin, then the valve must first be closed, and remain closed until the recycled water is pumped into the golf course lakes. The basin must dry throughout the summer before the valve may be opened again. Each date the drainage valve is closed or opened, and each date that discharge to the basin commences and terminates, shall be recorded in a logbook. The logbook shall be made available upon Executive Officer request.

14. Solids from the treatment process shall be disposed of at a facility approved by the Executive Officer.

C. GROUND WATER LIMITATIONS

1. The discharge shall not cause nitrate concentrations in the ground water downgradient of the disposal area to exceed 8 mg/L (as N).
2. The discharge shall not cause a significant increase of mineral constituent concentrations in underlying ground waters, as determined by comparison of samples collected from wells located upgradient and downgradient of the disposal area.^{BP}
3. The discharge shall not cause concentrations of chemicals and radionuclides in ground water to exceed limits set forth in Title 22, Chapter 15, Articles 4 and 5 of the California Code of Regulations.^{BP}

D. PROVISIONS

1. Order No. 89-03, "Wastewater Reclamation Requirements for Carmel Valley County Sanitation District, Producer of Recycled Water for Carmel Valley County," adopted by the Board on March 10, 1989 is hereby rescinded.
2. Producer shall comply with "Monitoring and Reporting Program No. 01-083," as specified by the Executive Officer.

3. Producer shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January 1984. For purpose of this Order, the words "disposal ponds" in the Standard Provisions shall mean the golf course lakes containing recycled water.

4. Pursuant to Title 23, Chapter 3, Subchapter 9, of the California Code of Regulations, the Producer must submit a written report to the Executive Officer no later than May 17, 2011, addressing:

- a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and,
- b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.

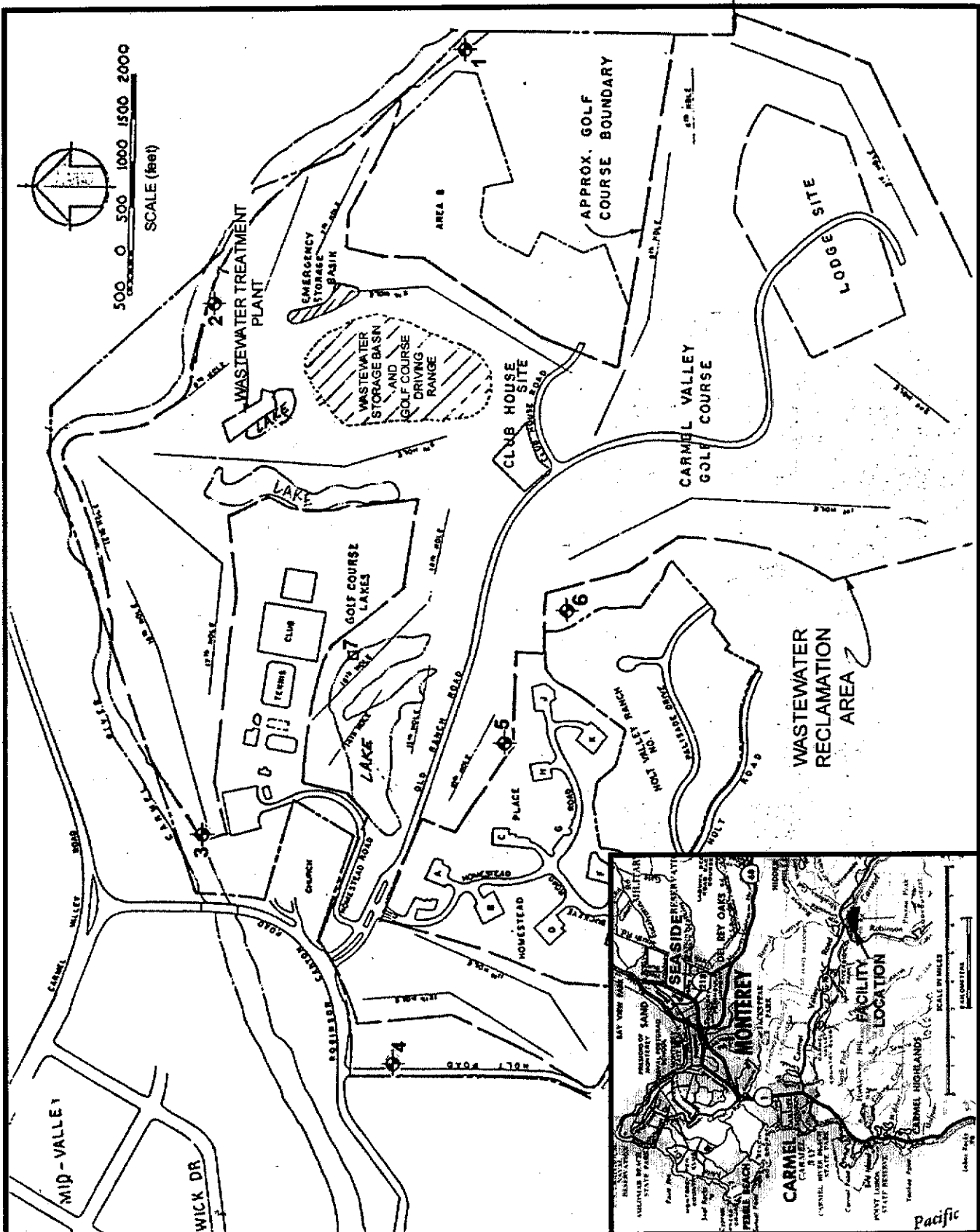
5. Producer shall comply with the following time schedule (See Paragraph "C.4" of the "Standard Provisions and Reporting Requirements"):

Task	Completion Date	Report Due
a. Install and operate a new influent pump station;	December 1, 2001	December 15, 2001
b. Submit an updated Operations and Maintenance Manual that includes a Sampling Procedures and Monitoring Plan;	--	April 1, 2002
c. Submit an engineering report that demonstrates Facility compliance with General Requirements of Design and Reliability Requirements, as found in Title 22, Division 4, Chapter 3, Articles 8 and 10, of the California Code of Regulations	--	April 1, 2002
d. Install and operate a drainage system that captures runoff from the solids disposal bin and returns the runoff to the Facility's influent (at the headworks).	March 18, 2002	April 1, 2002

I, **Roger W. Briggs**, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on September 14, 2001.


Executive Officer

S:\WB\Coastal Watershed\Staff\MThompson\Regulated Facilities\WDR\Carmel Valley Ranch\Adopted Orders 01-083 and 01-084\CVR WRR (Producer) Order No. 01-083.doc



**ATTACHMENT "A" - WRR ORDER NO. 01-083
 MONTEREY COUNTY SANITARY DISTRICT NO. 388, CARMEL VALLEY RANCH
 VICINITY AND FACILITY MAP**



California Regional Water Quality Control Board

Central Coast Region



Winston H. Hickox
Secretary for
Environmental
Protection

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Gray Davis
Governor

September 21, 2001

Mr. Joe Cochran
Monterey County Department of Public Works
312 E. Alisal Street
Salinas, CA 93901-4371

Ed Waggoner
Carmel Valley Ranch WWTP
One Old Ranch Road
Carmel, CA 93923

Martin Nicholson
Chris Dalhamer
Carmel Valley Ranch
One Old Ranch Road
Carmel, CA 93923

Dear Messrs. Cochran, Waggoner, Nicholson, and Dalhamer:

WATER RECYCLING REQUIREMENTS FOR MONTEREY COUNTY SANITATION DISTRICT NO. 388, CARMEL VALLEY RANCH ZONE NO. 1, PRODUCER OF RECYCLED WATER, AND CARMEL VALLEY RANCH GOLF COURSE, USER OF RECYCLED WATER, MONTEREY COUNTY

Enclosed is Order No. 01-083, Water Recycling Requirements for Monterey County Sanitation District No. 388, Carmel Valley Ranch Zone No. 1, Producer of Recycled Water, and Order No. 01-084, Water Recycling Requirements for Carmel Valley Ranch Golf Course, User of Recycled Water. Orders 01-083 and 01-084 were adopted by the Regional Water Quality Control Board at the September 14, 2001 meeting in Salinas. Please implement the requirements contained in these Order's immediately.

If you have questions, please contact **Matt Thompson at (805) 549-3159** or Gerhardt Hubner at (805) 542-4647.

Sincerely,

Roger W. Briggs
Executive Officer

Enclosures:

1. Water Recycling Requirements Orders 01-083 and 01-084;
2. Monitoring and Reporting Programs 01-083 and 01-084

California Environmental Protection Agency



Recycled Paper

cc:

Steve Setoodeh
California Department of Health Services
1 Lower Ragsdale, Building 1, Suite 120
Monterey, CA 93940

Monterey County Health Department
Environmental Health
1270 Natividad Road
Salinas, CA 93906

Cal-Am Water Company
P.O. Box 951
50 Ragsdale Drive, Suite 100
Monterey, CA 93940

Carmel Valley Property Owners Association
P.O. Box 157
Carmel Valley, CA 93924

S:\WB\Coastal Watershed\Staff\MThompson\Regulated
Facilities\WDR\Carmel Valley Ranch\Adopted Orders 01-
083 and 01-084\CVRTransmittal Order Nos. 01-083 and
01-084.doc



**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION**

MONITORING AND REPORTING PROGRAM NO. 01-083

Waste Discharger Identification No. 3 271018001

For

**MONTEREY COUNTY SANITATION DISTRICT NO. 388
CARMEL VALLEY RANCH ZONE NO. 1
PRODUCER OF RECYCLED WATER
FOR
CARMEL VALLEY RANCH GOLF COURSE
Monterey County**

INFLUENT MONITORING

Flows shall be measured and representative samples of the treatment plant influent shall be collected and analyzed as follows:

Constituent	Units	Type of Sample	Minimum Sampling and Analyzing Frequency
Flow Volume	Gallons	Metered	Continuous
Maximum Daily Flow	Gal/day	Calculated	Monthly
Mean Daily Flow	Gal/day	Calculated	Monthly
BOD ₅	mg/L	8-hr Composite	Monthly
Total Suspended Solids	mg/L	8-hr Composite	Monthly
pH	--	Grab	Monthly
Oil and Grease	mg/L	Grab	Monthly

RECYCLED WATER MONITORING

Flow shall be measured and representative samples of the treatment plant effluent shall be collected and analyzed as follows:

Constituent	Units	Type of Sample	Minimum Sampling and Analyzing Frequency
Daily Flow Volume	Gallons	Metered	Daily
Maximum Daily Flow	Gal/day	Calculated	Monthly
Mean Daily Flow	Gal/day	Calculated	Monthly
Total Coliform Organisms	MPN/100ml	Grab	Daily
Turbidity ¹	NTU	Metered	Continuous
Total Chlorine Residual	mg/L	Metered	Continuous

¹ Disinfected tertiary recycled water shall be continuously sampled for turbidity using a continuous turbidity meter and recorder following filtration. Compliance with the 2 NTU daily average limitation shall be determined by averaging the recorded turbidity levels at a minimum of four-hour intervals over a 24-hour period. Compliance with the 5 NTU limitation shall be determined using the recorded turbidity levels taken at intervals of no more than 1.2-hours over a 24-hour period. Should the continuous turbidity meter and recorder fail, grab sampling at a minimum frequency of 1.2-hours may be substituted for a period of up to 24-hours.

Constituent	Units	Type of Sample	Minimum Sampling and Analyzing Frequency
pH	Units	Grab	Weekly
BOD ₅	mg/L	8-hr Composite	Weekly
Total Suspended Solids	mg/L	8-hr Composite	Weekly
Total Dissolved Solids	mg/L	8-hr Composite	Quarterly (Jan, Apr, July, & Oct)
Chloride	mg/L	8-hr Composite	Quarterly (Jan, Apr, July, & Oct)
Sodium	mg/L	8-hr Composite	Quarterly (Jan, Apr, July, & Oct)
Total Nitrogen (Kjeldahl + Nitrate+ Nitrite)	mg/L	Grab	Quarterly (Jan, Apr, July, & Oct)

REPORTING

Quarterly monitoring reports shall be submitted within 30 days of the end of each quarter. For example, first quarter sampling results shall be submitted by April 30 of each year. In reporting the monitoring data, the discharger shall arrange the data in tabular form so the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that clearly illustrates compliance with water recycling requirements.

Annual monitoring reports shall be submitted by January 30 of each year, in accordance with Standard Provisions and Monitoring Requirements. Annual reports shall evaluate trends and summarize the previous year's data. In addition to the paper copy of the annual report, annual data shall be submitted electronically by 3.5" floppy diskette or electronic mail, preferably in Microsoft Excel file format.

ORDERED BY


Executive Officer

9-21-01

Date